Optimizitation and Validation of Okafor Reduced Reaction Mechanism

UROTHIYIL Aahil Ashraf (1TE23533T)

Department of Mechanical Engineering
Reactive Gas Dynamics Laboratory

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Traditional sources of energy rely on combustion of fossil fuels that produce carbon emissions which have adverse effects on the environment:

- Global Warming
- Climate Change

New carbon-free alternative fuels are needed to prevent exacerbating the effects of carbon pollutants

- The Japanese government seeks to reduce total greenhouse gas emissions by 46% before 2030
- By 2050, Japan envisions its cities to be carbon-neutral

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Ammonia, NH_3, is gaining interest among researchers for its potential to meet energy needs in a variety of applications.

Ammonia is a suitable fuel for combustion application because

- Ammonia combustion produces no CO_2
- Ammonia has a high energy density — both gravimetric and volumetry energy densities
- Easier to store than hydrogen
- Infrastructure to produce and transport ammonia already exists

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Drawbacks of using ammonia as a combustion fuel:

- Laminar burning velocity of ammonia is slower than methane (CH_4) and hydrogen (H_2)
- Ammonia has a narrow flammability range
- High NO_x emissions

To better study ammonia combustion, numerical simulations with reaction mechanisms are used.

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Abe Mechanism is developed based on Okafor Reduced Reaction Mechanism[1] to improve laminar burning velocity predictions for NH_3/H_2/Air flames.

- Introduces N_2H_x reaction pathways for more detailed $NH_2 \rightarrow NH$ reactions
- Recognises the importance of $NNH \leftrightarrow N_2 + H$ in reaction zone of the flame
- Updated constants for Arrhenius function for reactions

[1] Ekenechukwu Chijioke Okafor et al. "Measurement and Modelling of the Laminar Burning Velocity of Methane-Ammonia-Air Flames at High Pressures Using a Reduced Reaction Mechanism". In: *Combustion and Flame* 204 (June 1, 2019), pp. 162–175. ISSN: 0010-2180. DOI: 10.1016/j.combustflame.2019.03.008.

Objectives

- Numerically simulate flames with Abe mechanism at pressures 1, 3 and 5 atm for lean, stoichiometric and rich equivalence ratios.
  - Flame 1: $NH_3/H_2/Air$
  - Flame 2: $CH_4/NH_3/Air$
- Properties evaluated:
  - Laminar burning velocity
  - Species profiles of NO, NH_3, CO_2
  - Intermediate species profiles of OH, NNH, NH
- Validate by comparing numerically simulated values to experimental data from literature
- Optimize reaction mechanism to fit experimental data for all conditions above satisfactorily

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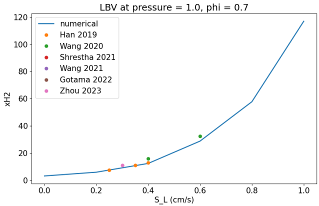
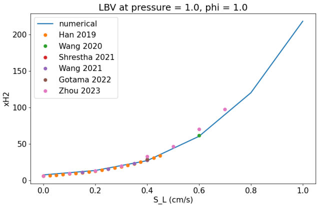
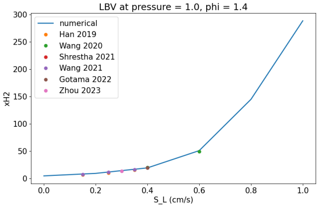
Laminar burning velocity of $NH_3/H_2/Air$ flame at 1atm of pressure



(a) lean equivalence ratio



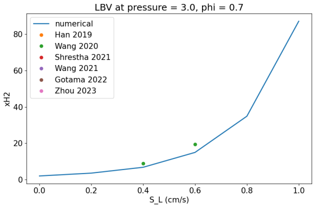
(b) stoichiometric equivalence ratio



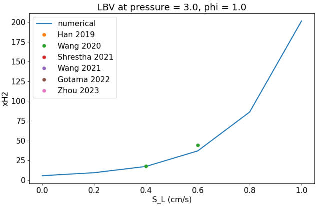
(c) rich equivalence ratio

Validating Abe Mechanism

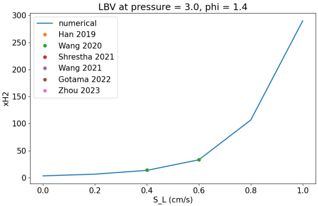
Laminar burning velocity of $NH_3/H_2/Air$ flame at 3atm of pressure



(a) lean equivalence ratio

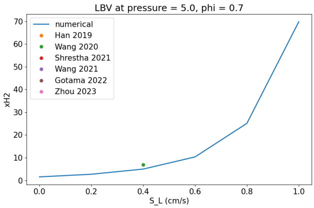


(b) stoichiometric equivalence ratio

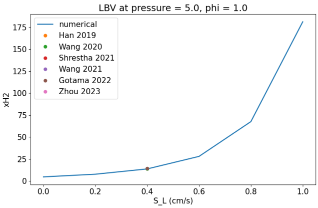


(c) rich equivalence ratio

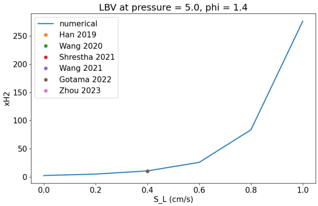
Laminar burning velocity of NH_3/H_2/Air flame at 5atm of pressure



(a) lean equivalence ratio



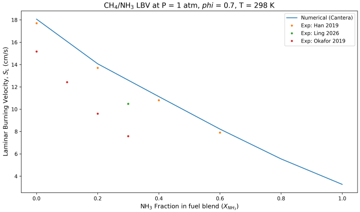
(b) stoichiometric equivalence ratio



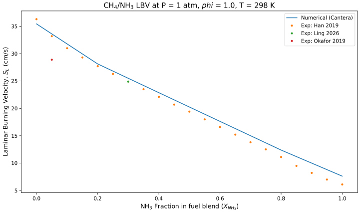
(c) rich equivalence ratio

Validating Abe Mechanism

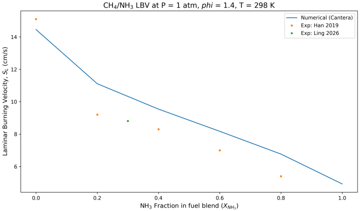
Laminar burning velocity of CH_4/NH_3/Air flame at 1atm of pressure



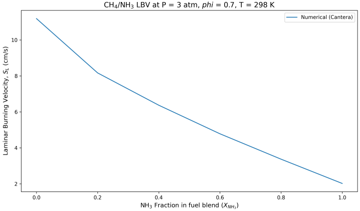
(a) lean equivalence ratio



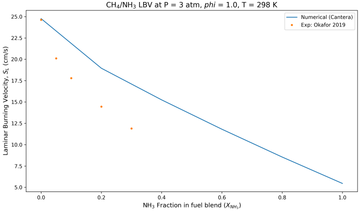
(b) stoichiometric equivalence ratio



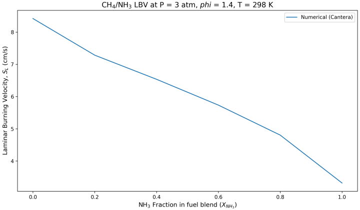
Laminar burning velocity of CH_4/NH_3/Air flame at 3atm of pressure



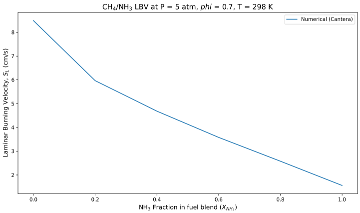
(a) lean equivalence ratio



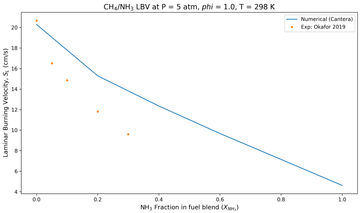
(b) stoichiometric equivalence ratio



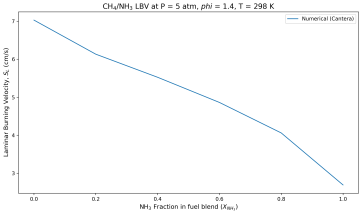
Laminar burning velocity of CH_4/NH_3/Air flame at 5atm of pressure



(a) lean equivalence ratio



(b) stoichiometric equivalence ratio



Remaining Objectives

- Numerically evaluate species profiles with Plug Flow Reactor
  - NO
  - NH_3
  - CO_2
- Numerically evaluate intermediate species profiles with Stagnation Flame Model
  - OH
  - NH
- Optimize reaction mechanism to fit experimental data for all conditions above satisfactorily

Gabriel J. Gotama et al. "Measurement of the Laminar Burning Velocity and Kinetics Study of the Importance of the Hydrogen Recovery Mechanism of Ammonia/Hydrogen/Air Premixed Flames". In: *Combustion and Flame* 236 (Feb. 1, 2022), p. 111753. ISSN: 0010-2180. DOI: 10.1016/j.combustflame.2021.111753. URL: https://www.sciencedirect.com/science/article/pii/S001021802100496X (visited on 05/19/2026)

Krishna Prasad Shrestha et al. "An Experimental and Modeling Study of Ammonia with Enriched Oxygen Content and Ammonia/Hydrogen Laminar Flame Speed at Elevated Pressure and Temperature". In: *Proceedings of the Combustion Institute* 38.2 (Jan. 1, 2021), pp. 2163–2174. ISSN: 1540-7489. DOI: 10.1016/j.proci.2020.06.197. URL: https://www.sciencedirect.com/science/article/pii/S1540748920302881 (visited on 05/19/2026)

Zhihua Wang et al. "Experimental and Kinetic Study on the Laminar Burning Velocities of NH3 Mixing with CH3OH and C2H5OH in Premixed Flames". In: *Combustion and Flame* 229 (July 1, 2021), p. 111392. ISSN: 0010-2180. DOI: 10.1016/j.combustflame.2021.02.038. URL: https://www.sciencedirect.com/science/article/pii/S0010218021001073 (visited on 05/19/2026)

Shixing Wang et al. "Experimental Study and Kinetic Analysis of the Laminar Burning Velocity of NH3/Syngas/Air, NH3/CO/Air and NH3/H2/Air Premixed Flames at Elevated Pressures". In: *Combustion and Flame* 221 (Nov. 1, 2020), pp. 270–287. ISSN: 0010-2180. DOI: 10.1016/j.combustflame.2020.08.004. URL: https://www.sciencedirect.com/science/article/pii/S0010218020303199 (visited on 05/19/2026)

Shangkun Zhou et al. "An Experimental and Kinetic Modeling Study on NH3/Air, NH3/H2/Air, NH3/CO/Air, and NH3/CH4/Air Premixed Laminar Flames at Elevated Temperature". In: *Combustion and Flame* 248 (Feb. 1, 2023), p. 112536. ISSN: 0010-2180. DOI: 10.1016/j.combustflame.2022.112536. URL: https://www.sciencedirect.com/science/article/pii/S0010218022005454 (visited on 05/19/2026)

Xinlu Han et al. "Experimental and Kinetic Modeling Study of Laminar Burning Velocities of NH3/Air, NH3/H2/Air, NH3/CO/Air and NH3/CH4/Air Premixed Flames". In: *Combustion and Flame* 206 (Aug. 1, 2019), pp. 214–226. ISSN: 0010-2180. DOI: 10.1016/j.combustflame.2019.05.003. URL: https://www.sciencedirect.com/science/article/pii/S0010218019302068 (visited on 05/19/2026)

Ekenechukwu Chijioke Okafor et al. "Measurement and Modelling of the Laminar Burning Velocity of Methane-Ammonia-Air Flames at High Pressures Using a Reduced Reaction Mechanism". In: *Combustion and Flame* 204 (June 1, 2019), pp. 162–175. ISSN: 0010-2180. DOI: 10.1016/j.combustflame.2019.03.008

Zhongqian Ling et al. "Experimental and Kinetic Modeling Study of the Laminar Burning Velocity of Pre-Dissociated NH3 + CH4 Flames". In: *Journal of the Energy Institute* 126 (June 1, 2026), p. 102503. ISSN: 1743-9671. DOI: 10.1016/j.joei.2026.102503. URL: https://www.sciencedirect.com/science/article/pii/S1743967126000632 (visited on 05/19/2026)